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Abiotic degradation of methylmercury mediated by hydroxyl radicals generated during ferrous oxidation in rice paddy soils

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ABSTRACT

Rice consumption is a primary exposure pathway for neurotoxic methylmercury (MeHg), with paddy soils acting as critical hotspots for net MeHg accumulation. Although microbial and photolytic demethylation are well-documented, dark abiotic demethylation mechanisms within light-impenetrable soil matrices remain poorly understood. This study investigated the dark abiotic MeHg degradation driven by hydroxyl radicals (·OH) generated during the oxygenation of different ferrous species: surface adsorbed Fe(II) on ferric oxides (≡Fe_3O_4-Fe(II)), ferrous sulfide (FeS), and ferrous disulfide (FeS_2). Upon air exposure, substantial ·OH accumulation occurred during the oxidation of ≡Fe_3O_4-Fe(II) (428.4 ± 5.82 μM) and FeS (561.9 ± 26.0 μM) systems over 72 h, whereas low-spin FeS_2 remained inert under our experimental conditions. In autoclaved paddy soil microcosms under oxic conditions, this oxidation cascade led to a significant dark demethylation of 35.45% (≡Fe_3O_4-Fe(II)) and 27.22% (FeS) of spiked MeHg over 48 h, while anoxic controls remained stable. Mechanistic analyses revealed divergent interfacial electron transfer pathways: surface passivation by amorphous Fe(III) limited the long-term reactivity of ≡Fe_3O_4-Fe(II), whereas the destruction of the FeS lattice and potential generated core–shell structure sustained internal electron pumping and prolonged MeHg demethylation (maximum MeHg demethylation rate constant of 0.192 h^{-1}). These findings elucidate a previously overlooked dark abiotic MeHg sink in redox-fluctuating rice paddy soils, demonstrating that Fe(II)-mediated Fenton-like reactions under alternating redox conditions can mitigate MeHg accumulation, a process that has not been accounted for in current Hg transport models.

1. Introduction

Mercury (Hg) is a ubiquitous global heavy metal pollutant; its organic form, methylmercury (MeHg), exhibits potent neurotoxicity and readily crosses the human blood–brain barrier (Clarkson, 1997). While natural wetlands and marine environments have historically been the focal points of MeHg cycling research (St. Louis et al., 2004; Poulin et al., 2016; Skyllberg et al., 2021), periodically flooded agricultural systems, such as rice paddy soils, are now recognized as critical hotspots for MeHg production (Rothenberg and Feng, 2012; Zhao et al., 2016a). The unique agronomic management of these soils drives dynamic redox fluctuations, creating active biogeochemical interfaces that foster the activity of Hg-methylating microorganisms (Zhao et al., 2016a; Liu et al., 2018). However, the net accumulation of MeHg in these dynamic ecosystems is not governed by methylation alone. Concurrently, natural MeHg demethylation occurs within these soil matrices (Barkay and Gu, 2021). Understanding these degradation processes is fundamentally crucial, as the interplay between methylation and demethylation dictates the net MeHg budget and its subsequent transport across biogeochemical interfaces.

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Demethylation of MeHg is driven by both biotic and abiotic processes (Barkay and Gu, 2021). Biotically driven demethylation proceeds predominantly via oxidative, reductive, and methanotrophic pathways (Barkay and Gu, 2021). Using $^{13}CH_3Hg^+$-DNA stable-isotope probing, Zhou et al. (2025) identified *Pseudarthrobacter*, *Methylophilaceae* (MM2), and *Dechloromonas* as active MeHg-demethylating taxa in rice paddy soils. Parallel to biotic processes, abiotic pathways also contribute significantly to MeHg demethylation in natural environments. Current understanding of abiotic demethylation revolves largely around two well-documented mechanisms: 1) photo-demethylation (Hammerschmidt et al., 2010; Li et al., 2010; Sun et al., 2015; Han et al., 2017; Zhong et al., 2024) and 2) dark reactions mediated by sulfides (Craig and Bartlett, 1978; Kanzler et al., 2018). In sunlit surface waters, photo-demethylation dominates, degrading over 80% of MeHg (Sellers et al., 1996; Zhang et al., 2010). Although photo-demethylation in bulk paddy soils is generally negligible due to limited light penetration, it accounts for up to 82% of MeHg loss in the overlying water of paddy fields (Zhong et al., 2024). Under ultraviolet irradiation, dissolved organic matter (DOM) generates reactive oxygen species (ROS), such as hydroxyl radicals (·OH), singlet oxygen ($^1O_2$), triplet excited state DOM (^{3}DOM*), and superoxide radicals ($O_2^{\cdot-}$), which facilitate the cleavage of the C-Hg bond (Zhang et al., 2010; Qian et al., 2014; Tai et al., 2014; Barkay and Gu, 2021). Conversely, in dark environments, sulfides produced by sulfate-reducing bacteria (SRB) can sequester MeHg via conversion to insoluble HgS (Craig and Bartlett, 1978; Kanzler et al., 2018). Additional pathways, such as selenium-dependent intracellular demethylation in biota (Khan and Wang, 2010; Wang et al., 2017) and $^1O_2$-mediated *in vivo* MeHg cleavage (Tang et al., 2024), have also been reported. Notably, our prior work documented the loss of spiked $Me^{198}Hg$ tracer in autoclaved rice paddy slurries under dark, anoxic incubation conditions (Liu et al., 2023), indicating the presence of a light- and sulfide-independent abiotic demethylation pathway. More recently, Xie et al. (2024) demonstrated that ferrous (Fe(II))-bearing minerals mediate the demethylation of MeHg to Hg^0 in dark, aqueous systems. However, whether this reactive ·OH-driven degradation can survive within complex soil matrices remains unknown. Accordingly, the activity and contribution of this Fe(II)-driven abiotic pathway to MeHg demethylation in paddy soils remain poorly understood.

In rice paddy soils, Fe is one of the most abundant redox-active elements (Kögel-knabner et al., 2010). Frequent flooding-drainage cycles create dynamic redox conditions that promote the accumulation of diverse Fe(II) species, such as Fe sulfides (FeS and FeS_2) and dissolved or adsorbed Fe(II), etc (Rickard and Luther, 2007; Kögel-knabner et al., 2010; Kappler et al., 2021). During episodic oxygen intrusion events, the abiotic oxidation of these redox-sensitive Fe(II) species triggers Fenton-like reactions. Even in dark environments, this process leads to the generation of highly reactive hydroxyl radicals (·OH), alongside intermediates including $O_2^{\cdot-}$ and hydrogen peroxide (H_2O_2) (Melton et al., 2014; Kappler et al., 2021). Extensive research confirms that these Fe(II) oxygenation-derived ROS acts as powerful, non-selective oxidants, profoundly influencing the biogeochemical cycling, mobility, and transformation of heavy metal and non-metallic contaminants (Meng et al., 2024; Zhang et al., 2025; Tan et al., 2026), as well as the degradation of organic pollutants (Chen et al., 2022; Yu et al., 2025). Li et al. (2025) reported that ·OH generated via FeS oxidation promotes the dissolution of mercuric sulfide nanoparticles, thereby releasing Hg(II). Given the strong oxidative capacity of ·OH and the pervasive nature of Fe(II) oxidation during redox fluctuations, it is highly plausible that ·OH generated from this dark abiotic process could act as a MeHg scavenger in rice paddy soils. However, due to the distinct interfacial electron transfer behaviors of different Fe(II) species in paddy soils, mechanisms underlying ·OH production and subsequent MeHg demethylation may differ.

To bridge this knowledge gap, this study selected representative Fe (II) species distributed in paddy soils, including surface adsorbed Fe(II) on ferric oxides (≡Fe_3O_4-Fe(II)), ferrous sulfide (FeS), and ferrous disulfide (FeS_2) to investigate the role of Fe(II) oxidation-induced ·OH in abiotic MeHg demethylation within redox fluctuated rice paddy soil. By elucidating the mechanisms governing this dark-abiotic reaction, this research aims to provide a novel perspective on the biogeochemical cycling of MeHg and also provides a crucial mechanistic framework for understanding the coupled biogeochemical cycling of heavy metals across Earth's critical zones.

2. Materials and methods

2.1. Preparation of different ferrous species

Paddy soils are typical Fe-rich environments where frequent flooding-drainage cycles create dynamic redox conditions, driving the extensive accumulation of diverse ferrous species, such as Fe sulfides and surface-adsorbed Fe(II) (Kögel-Knabner et al., 2010). Accordingly, three typical Fe(II) species were selected including ≡Fe_3O_4-Fe(II), FeS, and FeS_2 (Rickard and Luther, 2007; Huang et al., 2021; Wang et al., 2026). ≡Fe_3O_4-Fe(II) (Fe(Fe_3O_4): Fe(II) = 1:1) was prepared by adding $FeSO_4$ solution into magnetite (>97% purity, trace metals basis, 50 ~ 300 nm, Aladdin®, China) under dark. The prepared ≡Fe_3O_4-Fe(II) was equilibrated in the glovebox for overnight. FeS was prepared by combining Na_2S and $FeSO_4$ solutions and equilibrated overnight. FeS_2 was purchased from Macklin® (99.9% purity, trace metals basis) and grounded through 200 nylon mesh (74 μm) (Meng et al., 2024; Xie et al., 2024). All the Fe(II) species were prepared and equilibrated in aqueous buffer (pH 7.5) with a Fe(II) concentration of 1 g Fe(II) L^{-1}. Deoxygenated deionized water (DDW) and an O_2 free glovebox (5% H_2, 95% N_2, YQXII, Yuejin, China) was used for the preparation and equilibrium to avoid oxidation. The prepared Fe(II) species were characterized and confirmed by X-ray diffraction (XRD, SmartLab, Rigaku, Japan) and scanning electron microscope-energy dispersive X-ray spectroscopy (SEM-EDS, GeminiSEM360, ZEISS, USA) (Fig. S1).

2.2. Incubation experiment design

Abiotic oxidation experiments under ambient air exposure were conducted in three independent incubation series (Fig. S2).

Incubation 1: Fe(II) oxidation and ·OH production kinetics. Pre-characterized ≡Fe_3O_4-Fe(II), FeS, and FeS_2 suspensions in amber vials were exposed to ambient air outside the glovebox. All vials were covered with oxygen-permeating membranes on the top to prevent particulate contamination. The oxidation process proceeded on a horizontal orbital shaker (100 rpm) to ensure consistent gas–solid contact. For each Fe(II) species, three replicates were destructively sampled at 0, 0.5, 1, 2, 3, 5, 8, 12, 16, 24, 48, and 72 h. Samples at t = 0h were collected inside the glovebox to capture the initial redox state, while all subsequent time points were sampled under ambient air exposure to reflect the oxidation process. The ≡Fe_3O_4-Fe(II) oxidation experiment involved three treatments: 1) ≡Fe_3O_4-Fe(II), 2) pristine Fe_3O_4 (without adsorbed Fe(II)), and 3) a blank (buffer solution without Fe(II) or Fe(III)). Two treatments were included for both FeS and FeS_2 experiment (i.e., FeS/FeS_2 and Fe (II) free blank). During subsampling, aqueous and solid phases were separated using centrifugation. In the aqueous samples, concentrations of Fe(II), Fe(III), and ·OH were determined. For solid-phase samples, sequential extraction was performed using 20 mL of 0.5 M HCl to remove surface-adsorbed Fe, followed by 20 mL of 5 M HCl to extract structural Fe (Chen et al., 2022). Because the mass of the solid phase varied during oxidation, Fe concentrations are reported on a volumetric basis with a fixed extraction agent volume of 20 mL.

Incubation 2: MeHg demethylation and the reproducibility of Fe(II) oxidation and ·OH production. In this incubation, FeS_2 was omitted due to the lack of detectable ·OH production during oxidation in this study. Prior to oxidation, a MeHg standard (93160a, TMRM®, China) was spiked into the Fe(II)-containing vials to achieve a final MeHg

concentration of 2.5 ng L^{-1} (paddy porewater relevant level in Hg contaminated region, 0.62 ~ 4.7 ng L^{-1}) (Zhao et al., 2016b). Based on the results of Incubation 1, the sampling time points were adjusted to 0, 0.5, 1, 2, 3, 8, 12, 24, and 48 h. The concentrations of MeHg and THg were determined in both the aqueous and solid phases. The remainder of the experimental design remained identical to that of Incubation 1 (Fig. S2). Furthermore, all chemicals and materials used in both Incubations 1 and 2 were autoclaved prior to use.

Incubation 3: MeHg demethylation under paddy soil incubation. Surface paddy soil (0–20 cm) was collected from paddy fields located in a rural area in Guiyang city, China, with elevated geogenic Hg levels ([THg] = 0.16 ± 0.01 mg kg^{-1}) in April 2022 (Huang et al., 2025). The collected soil was air-dried, cleared of rocks and plant residues, and sieved through a 2-mm nylon mesh. To eliminate biotic processes, 50 g of the prepared soil was placed into 100-mL borosilicate glass bottles and autoclaved. It should be noted that while autoclaving may partially alter soil physicochemical properties, it was selected over alternative methods like gamma irradiation, which may induce the generation of ROS from our previous work (Hao et al., 2024). Ferrous species (≡Fe_3O_4-Fe(II) and FeS) were added via solution (40 mL, 1 g Fe(II) L^{-1}) and evenly mixed into the soil. The prepared paddy slurry was equilibrated overnight in the glovebox. Prior to oxidation, a MeHg standard (93160a, TMRM®, China) was spiked into the slurry to achieve a final concentration of 4 μg kg^{-1} (paddy soil relevant level in Hg contaminated region, 0.76 ~ 6.2 μg kg^{-1}) (Zhao et al., 2016b). It is noted that the spiked MeHg concentration in Incubation 3 is higher than those in batch experiment (Incubation 2). This is to ensure the ·OH-driven MeHg demethylation signal could be tracked against the complexation effects of natural organic matter (Baptista-Salazar et al., 2022). Time series were resettled as 0, 1, 2, 4, 8, 24, and 48 h. A 48 h's incubation was selected due to the rapid demethylation of MeHg, and prolonging the incubation periods may result resuscitation of resilient microbes and subsequent inorganic Hg methylation. This timescale aligns with standard Hg tracer incubation protocols that are conventionally set at 6–48 h to minimize confounding biological interferences (Helmrich et al., 2022). Three replicates for each Fe(II) species at each sampling time were destructively collected. In addition to the air exposed treatment, anaerobic controls were also set up in a UV-irradiated glovebox (Fig. S2). Oxidation was proceeded outside the glovebox with gentle stirring of the slurry in each bottle using sterile Teflon-coated stir bars.

2.3. Measurements

Concentrations of aqueous Fe(II) and Fe(III) (Total Fe-Fe(II)) were determined using ferrozine method by a UV–vis spectrometer (UV-5100, METASH, China) at 562 nm (Viollier et al., 2000). Total Fe was determined after 10% NH_2OH·HCl (*w/v*) reduction. Solid phase Fe(II) and Fe(III) were determined after 0.5 M HCl (adsorbed Fe) and 5 M HCl (structural Fe) extraction and filtration (0.45 μm PES filter, JINTENG®, China). Ferrous species after oxidation were characterized by X-ray photoelectron spectroscopy (XPS) and Mössbauer spectrum. Specifically, the surface chemical states were characterized by XPS (ESCALAB Xi+, Thermo Fisher, USA) equipped with a monochromatic Al Kα radiation ($h\nu$ = 1486.6 eV). Operating parameters included a 12 kV voltage, 6 mA current, and a 400 μm spot size. High-resolution narrow scans of the Fe2p region were collected (pass energy: 10 eV, step size: 0.1 eV). Binding energies were calibrated by using the C1s peak at 284.8 eV. To further identify Fe species with different oxidation sates, ^{57}Fe Mössbauer spectroscopy was conducted. Briefly, the solid samples were finely ground into powder and analyzed using a Mössbauer spectrometer (W304, SEE Co., USA) at room temperature. A ^{57}Co/Rh radioactive source was employed for γ-ray emission, and the velocity scale was calibrated against a standard α-Fe foil. More details of measurement and key parameters from Mössbauer spectroscopy can be found in Text S1 and previous works (Liu et al., 2021; Qi et al., 2025).

Concentration of ·OH was determined by a chemical probe of benzoic acid ([BA] = 30 mM, probe-to-sample volume ratio of 1:2 (*v/v*)) by forming p-hydroxybenzoic acid (*p*-HBA) (Tong et al., 2016; Liao et al., 2019). This BA concentration is excessive to quantitatively capture the generated ·OH under the experimental conditions. Measurement for *p*-HBA was proceeded with an HPLC (1260, Agilent, USA) equipped with a UV–vis detector (detection wavelength of 255 nm). The mobile phase was a mixture of acetonitrile (HPLC grade) and 1% TFA/water (65:35, *v/v*), with a C_{18} column ($5C_{18}$-PAQ, COSMOSIL, Japan). In addition to quantitative analysis, ·OH was also detected by electron paramagnetic resonance (EPR) spectroscopy (EMXplus, Bruker, Germany). 5,5-dimethyl-1-pyrroline-N-oxide (DMPO, 200 μL) was used as the spin trap. Instrumental parameters are as below: microwave power, 20 mW; resonance frequency, 9.8 ~ 9.9 GHz; modulation frequency, 100 kHz; modulation amplitude, 1.0 G; and sweep time, 80 s.

Filtered aqueous THg was determined using a cold vapor atomic fluorescence spectroscopy (CVAFS, HXAM-HgAF, Hongxin, China) after BrCl oxidation, and $SnCl_2$ reduction (EPA 1631, 2002). Solid THg was digested by aqua regia (HCl/HNO_3 = 3:1 *v/v*) and determined by CVAFS (EPA 1631, 2002). Filtered aqueous MeHg was determined using a GC-cold vapor atomic fluorescence spectroscopy (GC-CVAFS, PGC-900, Hongxin, China) followed by $NaBEt_4$ ethylation, and Tenax® trap preconcentration (EPA 1630, 2001). Solid phase MeHg was extracted by HNO_3-$CuSO_4$-CH_2Cl_2 and determined by GC-CVAFS (Liang et al., 1996).

2.4. Data analysis, statistics and QA/QC

Concentration of ·OH was quantified as 5.87 times of the concentration of *p*-HBA (Liao et al., 2019). Soil mineral phases from XRD were retrieved by using MDI Jade 6 software (Materials Data Inc., USA). Peak fitting and raw spectra smoothing of XPS were conducted using Avantage Software with a Shirley baseline. The acquired Mössbauer spectra were then quantitatively analyzed and deconvoluted using the Recoil software (Ottawa, Canada) through Voigt-based fit (Liu et al., 2021). Key hyperfine parameters (i.e., isomer shift, quadrupole splitting, and hyperfine magnetic field, Text S1) were extracted for phase identification.

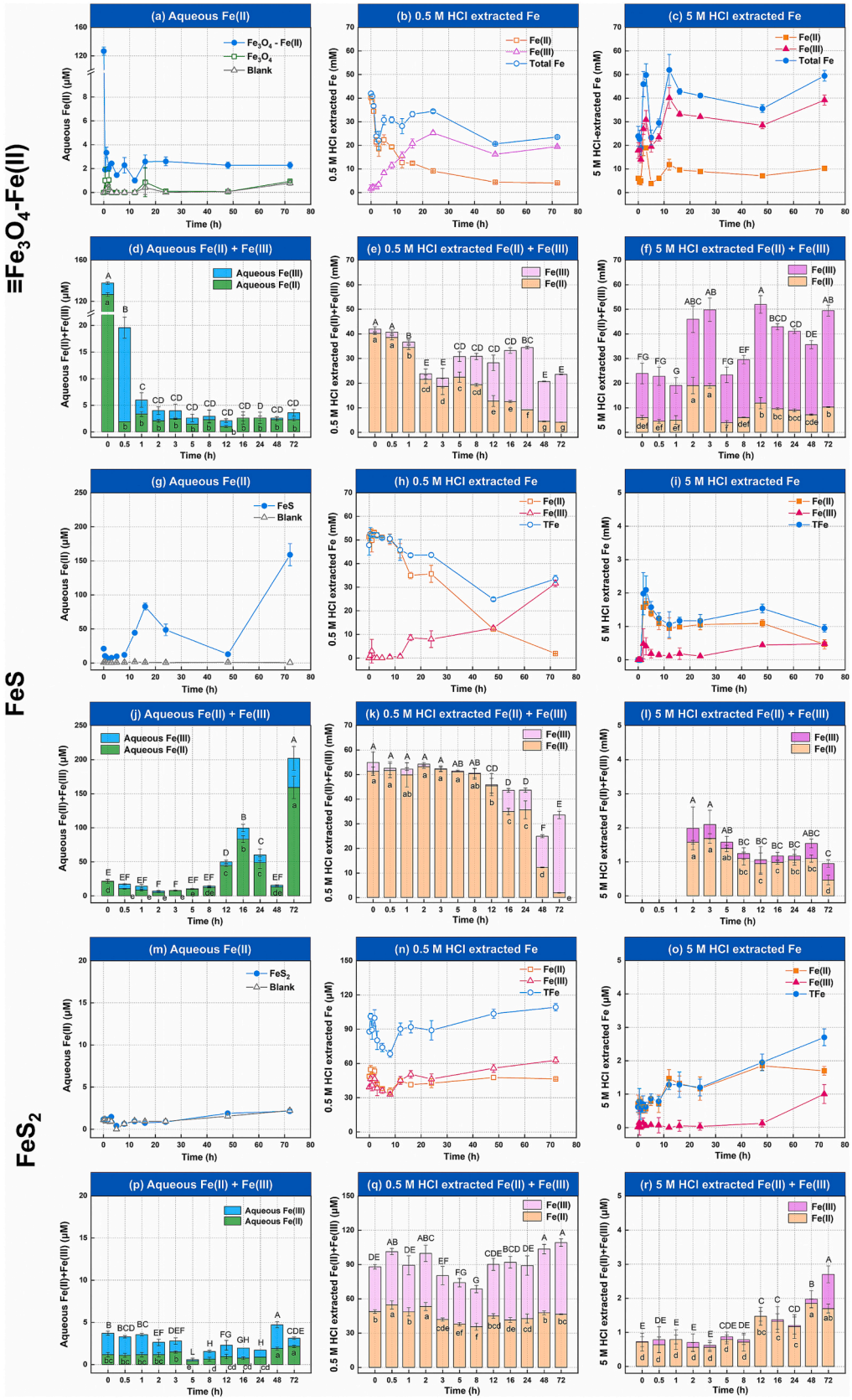
Statistics were performed using SPSS 27.0 (IBM, IL). Both parametric tests (independent samples *t* test, paired samples *t* test, and one-way ANOVA with Duncan's post hoc test) and nonparametric tests (Mann-Whitney *U* test and Kruskal-Wallis one-way ANOVA) were applied for the difference test. Statistical significance was set at $p < 0.05$.

Detection limit of *p*-HBA was 0.1 μM. Detection limit (3σ) for THg and MeHg are 0.06 ng L^{-1} (15 mL) and 0.006 ng L^{-1} (45 mL), respectively. During the measurement for incubation 1 and 2, diluted Fe(II) (Aladdin®, 100 μg mL^{-1} in 5% HCl), THg (GBW-083186, TMRM®, China) and MeHg (93160a, TMRM®, China) standards were applied for quality control. Recovery ranges of THg and MeHg are 86.65% ~107.83% (n = 26) and 88.02%~104.21% (n = 26), respectively. For the soil samples, certified reference materials of GSS-66 ([THg] = 0.19 ± 0.01 mg kg^{-1}) and CC580 (ERM®, [MeHg] = 0.075 ± 0.0037 mg kg^{-1}) were used. Recoveries for soil THg and MeHg are 89.23 ± 6.40% (n = 32) and 90.73 ± 5.07% (n = 24), respectively.

3. Results and discussions

3.1. Fe(II) oxidation

Abiotic oxidation process of ≡Fe_3O_4-Fe(II), FeS, and FeS_2 was monitored by Fe(II) and Fe(III) measurements in aqueous and solid phases (0.5 M HCl and 5 M HCl extracted) in Incubation 1. During the initial 0.5 h of ≡Fe_3O_4-Fe(II) oxidation, aqueous Fe(II) decreased rapidly, from 126.58 ± 5.67 μM to 1.91 ± 0.001 μM ($p < 0.05$, Fig. 1a), concomitant with a marked increase in aqueous Fe(III) (Fig. 1d), confirming rapid oxidative dissolution of aqueous Fe(II). However, aqueous



(caption on next page)

Fig. 1. Fe(II) and Fe(III) dynamics during ≡Fe_3O_4-Fe(II) (a-f), FeS (g-l), and FeS_2 (m-r) oxidation in aqueous and solid phases in Incubation 1. Panels (a), (g), (m) are the concentration of aqueous Fe(II); panels (d), (j), (p) are the sum of aqueous Fe(II) and Fe(III); panels (b), (h), (n) are the concentrations of Fe(II), Fe(III), and total Fe in solid phase extracted by 0.5 M HCl; panels (e), (k), (q) are the sum of 0.5 M HCl extracted Fe(II) and Fe(III); panels (c), (i), (o) are the concentrations of Fe(II), Fe (III), and total Fe in solid phase extracted by 5 M HCl; panels (f), (l), (r) are the sum of 5 M HCl extracted Fe(II) and Fe(III). Difference lowercase letters and capital letters in stacked bar plots indicate significant difference of Fe(II) and total Fe among time points, respectively ($p < 0.05$). The missing data in panel (l) is attributed to the complete dissolution caused by 0.5 M HCl extraction.

Fe(III) subsequently declined and stabilized between 0.5 and 1 h (Fig. 1d), indicating its removal from solution into the solid phase. In the solid phase extracted by 0.5 M HCl, Fe(II) concentration decreased continuously ($p < 0.05$), and the dominant Fe species shifted from Fe(II) to Fe(III) after 12 h of oxidation (Fig. 1b and e). In contrast, sequential extraction using 5 M HCl revealed that Fe(III) remained the dominant Fe species across all time points (Fig. 1c and f). These results collectively indicate that Fe(II) oxidation in ≡Fe_3O_4-Fe(II) occurs predominantly at the mineral surface. XPS and Mössbauer spectroscopic results (Fig. 2) are consistent with this interpretation. Specifically, XPS Fe2p spectra revealed a decrease in surface Fe(II) from 35.1% to 21.4% upon oxidation of ≡Fe_3O_4-Fe(II) (Fig. 2a and b). Concurrently, Mössbauer spectral fitting showed that the fraction of surface-adsorbed Fe(II) (Fe^{2+}_{ads}) declined from 22.5% to 17.7%, whereas the surface-adsorbed Fe(III) (Fe^{3+}_{ads}) fraction increased from 34.1% to 40.3% (Table S1). These surface-adsorbed Fe species (represented as doublets) were readily distinguished from the structural Fe in magnetite (characterized by magnetic sextets), consistent with the spectral deconvolution approach for surface-bound and precipitated Fe species established by Soltermann et al. (2014). In contrast, the proportions of structural Fe(II) and Fe(III) within the Fe_3O_4 lattice remained stable, shifting only marginally from 15.1% to 13.1% and from 28.4% to 28.9%, respectively (Table S1). This indicates that the bulk magnetite core was shielded from air oxidation. Notably, the quadrupole splitting (QS) value of the residual Fe^{2+}_{ads} expanded from 2.23 mm/s to 2.57 mm/s upon oxidation (Table S1). This QS increase reflects reduced local symmetry in the electric field gradient surrounding the remaining adsorbed Fe(II) nuclei, which is likely induced by localized surface distortion or the precipitation of secondary Fe(III) phases (Notini et al., 2023). Over 72 h, the overall oxidation extent of solid-phase Fe(II) (i.e., the sum of 0.5 M and 5 M HCl-extractable Fe(II)) in the ≡Fe_3O_4-Fe(II) system reached 69.0% (Fig. S3d). Moreover, upon oxidation of ≡Fe_3O_4-Fe(II), the proportion of 0.5 M HCl-extracted Fe out of the total solid Fe (0.5 M + 5 M HCl extractions) decreased significantly from 63.7% to 32.2% ($p < 0.05$, Fig. S3a), indicating incorporation of surface-adsorbed or loosely bound Fe into the structural lattice (Gorski et al., 2012). This structural incorporation, together with the Mössbauer-derived distortion in the coordination environment of residual Fe^{2+}_{ads} (i.e., increased QS value), provides evidence that adsorbed Fe(II) serves as an interfacial electron donor to the magnetite, thereby initiating coupled dissolution–recrystallization (Liu et al., 2021; Fang et al., 2023).

During FeS oxidation, aqueous Fe(II) concentrations fluctuated over time, peaking at 159.0 ± 16.3 μM after 72 h ($p < 0.05$; Fig. 1g and j). Fe (II) remained the dominant Fe species in the aqueous phase throughout the entire oxidation process (Fig. 1j). In the solid phase extracted with 0.5 M HCl Fe(II) was initially dominant, but Fe(III) became the predominant species by 48 h and reached a final concentration of 31.6 ± 1.44 mM (Fig. 1h and k). In contrast to ≡Fe_3O_4-Fe(II), Fe(II) remained the dominant Fe species in the 5 M HCl extractable fraction (Fig. 1i and l). Notably, more than 93.8% of total solid-phase Fe was extractable by 0.5 M HCl (Fig. S3b). Integrating quantification of Fe(II) and Fe(III) from both 0.5 M and 5 M HCl extractions revealed significant solid-phase Fe (III) accumulation beginning after 16 h of oxidation ($p < 0.05$, Fig. S3e). Over 72 h, the overall oxidation extent of solid-phase Fe(II) (i.e., the sum of 0.5 M and 5 M HCl-extractable Fe(II)) reached 95.4% in the FeS system (Fig. S3e). Through Mössbauer spectroscopy (Fig. 2g), the solid residue comprised 37.45% unreacted FeS, 43.23% newly formed Fe(III) phases, and a distinct Fe(II) component (19.32%) assigned to $FeSO_4$ (Table S1) after oxidation. The presence of $FeSO_4$ provides direct evidence that O_2 exposure triggers not only FeS lattice dissolution, but also concurrent oxidation of structural sulfide (HS^-) to sulfate (SO_4^{2-}) (Cheng et al., 2023). This inference is further supported by XPS S2p spectra,

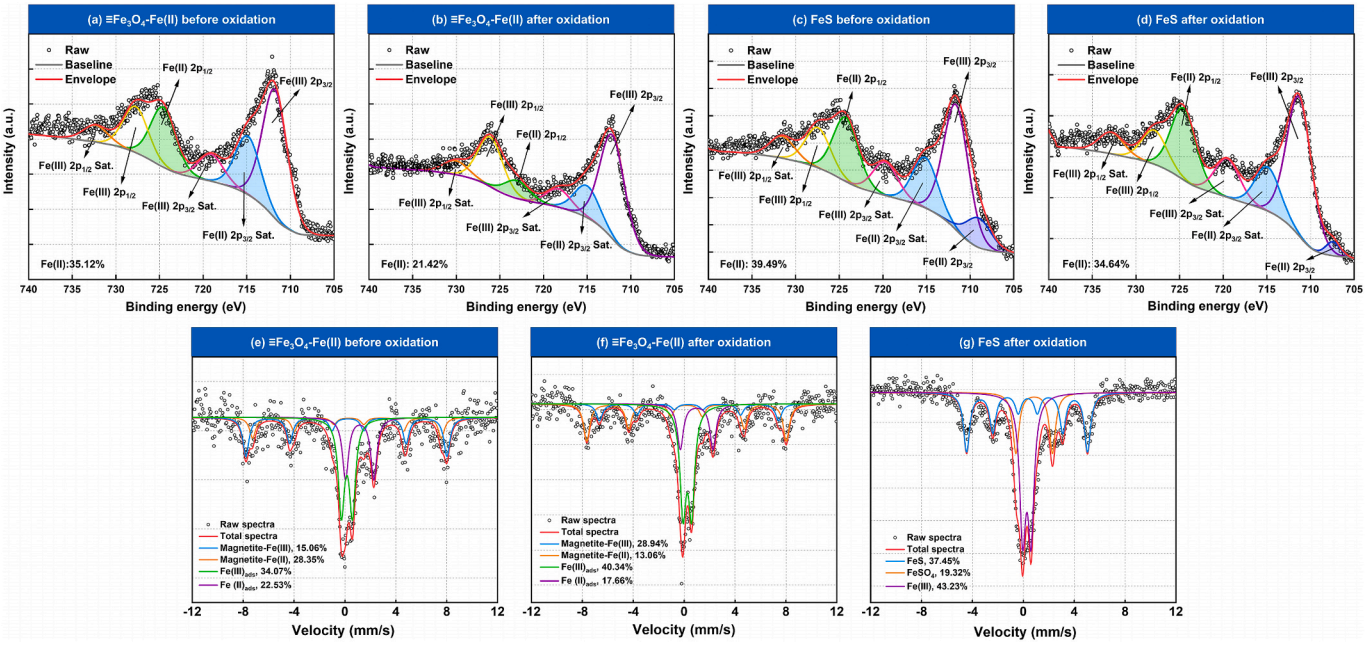


Fig. 2. Surface chemical states and Fe mineralogical transformation of ≡Fe_3O_4-Fe(II) and FeS during oxidation. Panels (a-d) are high-resolution Fe2p XPS spectra of ≡Fe_3O_4-Fe(II) and FeS before and after oxidation. Panels (e-g) ^{57}Fe Mössbauer spectra of the corresponding samples measured at 295 K. Red solid lines represent the fitting results for total spectra, and open circles denote the raw data. Percentages of each fitted Fe species are from Table S1.

which show a decrease in reduced sulfur (S(−II)) from 30.8% to 24.6% upon oxidation (Figs. S4a and b). This coupled, multi-element oxidation cascade explains why the vast majority of the transformed Fe products remained amorphous or poorly crystalline phases (Li et al., 2023) that are readily extractable by 0.5 M HCl (Fig. S3b).

For FeS_2, air-induced oxidation was negligible in both aqueous phase and solid-phases (Fig. 1m-1r). Relative stable Fe(II) and Fe(III) concentrations were observed in 0.5 M HCl extractable Fe (Fig. 1q). It is noted that, FeS is highly susceptible to air oxidation due to its reactive high-spin Fe(II) and metallic bonding. In contrast, under our short-term experimental conditions, FeS_2 remains inert because its Fe(II) exists in a stable low-spin closed-shell configuration that thermodynamically and kinetically hinders oxidation (Rickard and Luther, 2007). Although recent studies demonstrate that prolonged FeS_2 oxidation can eventually mediate ·OH generation (Zhao et al., 2022; Ma et al., 2026), this process was negligible within our incubation periods. Regarding 5 M HCl extractable Fe, both Fe(II) and Fe(III) concentrations increased significantly ($p < 0.05$, Fig. 1o and r), indicating that Fe(II) oxidation occurred only upon partial destabilization of the FeS_2 lattice induced Fe(II) releases. As demonstrated by Zhao et al. (2022), slow pyrite oxidation generates secondary Fe-bearing phases, including surface-associated (*in situ*) coatings and detached (*ex situ*) precipitates, many of which are acid-labile. Nevertheless, throughout the 72-h reaction period, the total concentration of HCl-extractable Fe remained in the micromolar (μM) range, approximately three orders of magnitude lower than in the ≡Fe_3O_4-Fe(II) and FeS systems. This confirms that the vast majority of Fe in FeS_2 is refractory to both 0.5 M and 5 M HCl extraction.

3.2. ·OH production

After 72 h of oxidation, ·OH accumulated to 428.4 ± 5.82 μM in the ≡Fe_3O_4-Fe(II) system and 561.9 ± 26.0 μM ·OH in the FeS system in Incubation 1 (Fig. 3a and b). In contrast, no ·OH was detected in blanks or in the Fe_3O_4-only control, confirming that ·OH production is specifically attributable to the oxidation of Fe(II) adsorbed on the surface of Fe_3O_4. Notably, no detectable ·OH signal was observed in the FeS_2 system (Fig. 3c), consistent with the negligible Fe(II) oxidation observed in Fe speciation analyses in this study (Fig. 1m-1r). Kinetic modeling, including pseudo-first-order model and pseudo-second-order model, was applied to elucidate the mechanistic basis of ·OH generation. Results showed that ·OH production in the ≡Fe_3O_4-Fe(II) system followed pseudo-first-order kinetic model ($R^2 = 0.996$) with a rate constant (k_1) of 0.11 h^{-1} and a modeled maximum production (C_{max}) of 418.96 μM (Fig. S5a). This C_{max} value aligned closely with the experimental

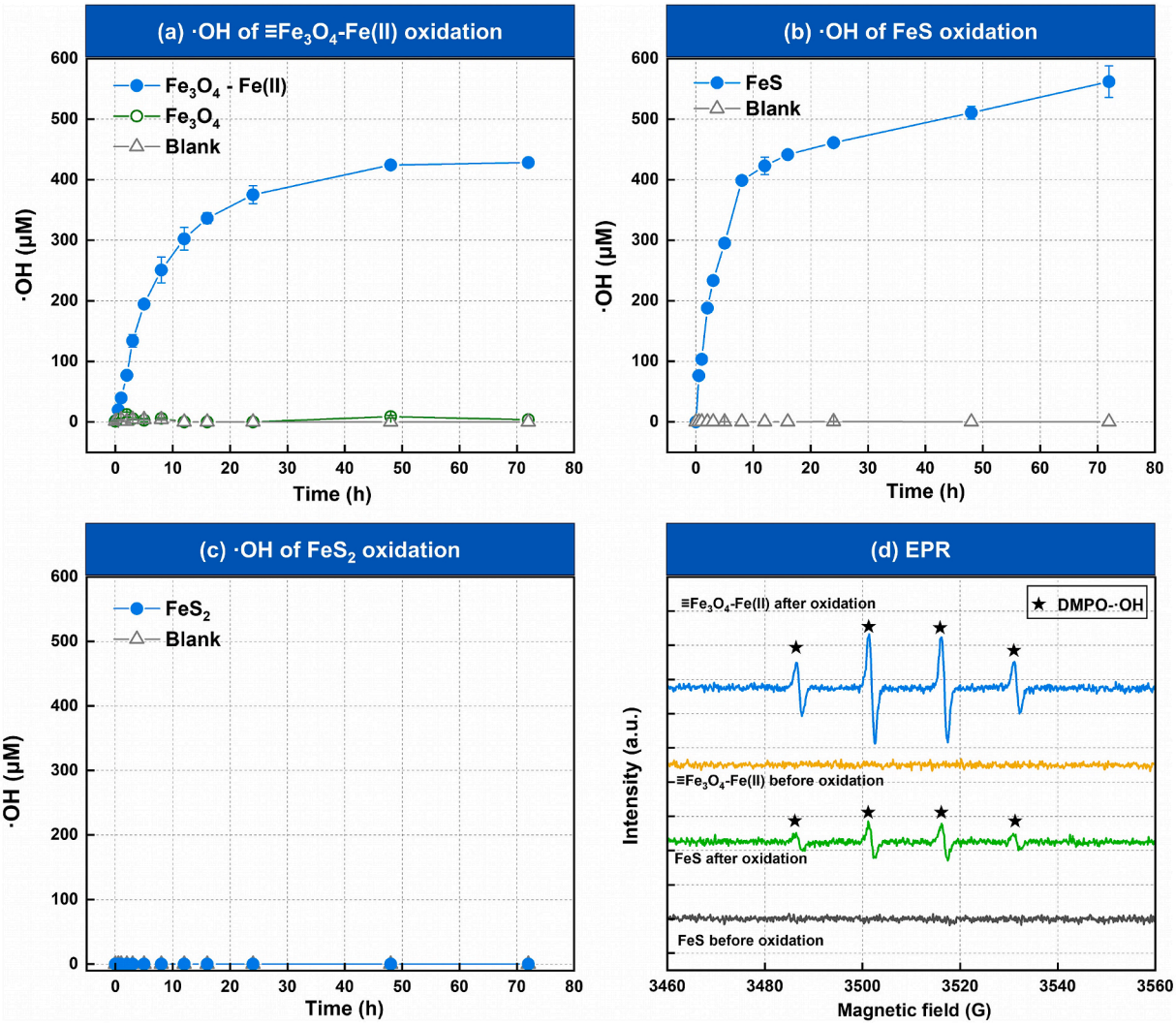


Fig. 3. Production of hydroxyl radicals (·OH) by ≡Fe_3O_4-Fe(II), FeS, and FeS_2 during oxidation in Incubation 1. Time-course measurements of ·OH concentration are shown for the (a) ≡Fe_3O_4-Fe(II), (b) FeS, and (c) FeS_2 systems, compared to respective blank controls. Panel (d) displays electron paramagnetic resonance (EPR) spectra of ≡Fe_3O_4-Fe(II) and FeS before and after oxidation. The four-line signature peak of the DMPO-·OH adduct is indicated by asterisks.

measurement (428.4 ± 5.82 μM at 72 h, Fig. 3a). This excellent fit indicates that the ·OH generation rate in this system is primarily governed by a single limiting factor: the density of available active Fe(II) sites on the surface (Kwan and Voelker, 2003). As the surface reactive Fe(II) is consumed and an amorphous Fe(III) passivation layer forms, interfacial electron transfer is increasingly impeded (Wang et al., 2016). In contrast, ·OH generation in the FeS system was better described by the pseudo-second-order model ($R^2 = 0.993$). This suggests a more complex interfacial reaction mechanism for FeS, which is likely controlled by chemisorption or dynamic interactions involving multiple reactive sites (Ho and Mckay, 1999). This differentiation is further substantiated by the relationships observed between cumulative ·OH production and Fe (II) pools (Fig. S6). Specifically, ≡Fe_3O_4-Fe(II) oxidation exhibited a linear fit (Fig. S6a), whereas FeS oxidation followed a nonlinear relationship (Fig. S6b), consistent with a multi-site-controlled oxidation of FeS. Moreover, although prior studies have reported significant linear correlations between cumulative production of ·OH and total Fe(II) (Cheng et al., 2016; Liao et al., 2019), our data show that cumulative ·OH accumulation was quantitatively predicted only by the 0.5 M HCl extractable Fe(II), rather than by aqueous Fe(II) or 5 M HCl extractable Fe(II) in both ≡Fe_3O_4-Fe(II) and FeS systems (Fig. S6). This selective dependence indicates that ·OH generation is predominantly mediated by the oxidation of surface-adsorbed or weakly-bound Fe(II).

To further verify the generation of ·OH during the oxidation process, electron paramagnetic resonance (EPR) spectroscopy was performed using DMPO as a spin-trapping agent. No EPR signals attributable to DMPO-·OH were detected in the pristine ≡Fe_3O_4-Fe(II) or FeS suspensions prior to oxidation (Fig. 3d). Following oxidation, both systems exhibited a characteristic four-line EPR signal with intensity ratio 1:2:2:1 and *g*-value of 2.002, which is the fingerprint of the DMPO-·OH spin adduct (Liao et al., 2019). This spectroscopic confirmation corroborates the benzoic acid probe results and confirms that ·OH is generated *in situ* via interfacial Fe(II) oxidation.

Production pathways of ·OH mediated Fe(II) abiotic oxidation is well-established through Fenton-like reactions by H_2O_2 disproportionation in following steps (Melton et al., 2014; Kappler et al., 2021):

$$Fe(II) + O_2 \rightarrow Fe(III) + O_2^{\cdot -} \quad (1)$$

$$Fe(II) + O_2^{\cdot -} + 2H^+ \rightarrow Fe(III) + H_2O_2 \quad (2)$$

$$Fe(II) + H_2O_2 + H^+ \rightarrow Fe(III) + \cdot OH + H_2O \quad (3)$$

Through density functional theory (DFT) calculations, Meng et al. (2024) suggests that surface oxygen vacancies and active crystal facets can effectively lower the thermodynamic energy barriers for the adsorption and activation of intermediates ($O_2^{\cdot -}$ and H_2O_2). In the context of our findings, these active surface sites likely serve as the primary coordination environments for the 0.5 M HCl-extractable Fe(II) fraction. Such favorable coordination enhances interfacial electron transfer from subsurface or structural Fe(II) to adsorbed oxidants, thereby rationalizing the sustained and robust ·OH generation observed in these heterogeneous Fenton-like systems.

3.3. MeHg demethylation

Batch incubations (Incubation 2, Fig. S2) were first employed to investigate the role of ·OH production in MeHg demethylation, followed by autoclaved soil microcosms (Incubation 3, Fig. S2) to validate this abiotic process in paddy soil matrices. During Incubation 2, the dynamics of Fe(II)/Fe(III) species across different phases were monitored (Figs. S7 and S8), confirming that the Fe(II) oxidation processes previously observed in Incubation 1 (Fig. 1) were reproducible. However, upon spiking with MeHg in Incubation 2, overall ·OH production noticeably decreased compared to the unspiked system in Incubation 1. The maximum ·OH concentrations only reached 363.0 ± 10.11 μM and 466.9 ± 8.53 μM in the ≡Fe_3O_4-Fe(II) and FeS systems, respectively (Fig. S9). These values account for 84.7% and 83.1% of the ·OH yields observed in the corresponding MeHg-free incubations (Fig. 3a and b). The observed reduction in apparent ·OH accumulation is likely driven by the rapid consumption of ·OH during MeHg demethylation, alongside competitive scavenging by its intermediates and potential matrix interferences. During the oxidation of ≡Fe_3O_4-Fe(II) in Incubation 2, decreases of MeHg in aqueous phases were observed after an initial increase (Fig. 4a). In solid phases, however, MeHg concentrations were not detected. During FeS oxidation in Incubation 2, MeHg was detectable only in the Fe-free control (i.e., only MeHg treatment), but remained undetectable in both the aqueous and solid phases (Fig. 4b). This suggests a rapid demethylation of MeHg likely occurred during the oxidation of FeS system. Crucially, differentiation in MeHg loss mentioned above coincides with the higher rate constants ($k_1 = 0.195$ h^{-1}, $k_2 = 0.00042$ μM h^{-1}, Fig. S5) of ·OH production in FeS system compared to the ≡Fe_3O_4-Fe(II) system ($k_1 = 0.11$ h^{-1}, $k_2 = 0.0003$ μM h^{-1}, Fig. S5).

Significant demethylation of MeHg was observed in both ≡Fe_3O_4-Fe (II) and FeS-amended oxidizing systems during Incubation 3 with autoclaved paddy soil (Fig. 4c). In contrast, MeHg concentrations in anoxic controls, defined as unoxidized systems spiked with MeHg and Fe (II) species, remained stable throughout the experimental period, exhibiting only minor fluctuations (Fig. 4c). This indicates that abiotic MeHg demethylation mediated by Fe(II) oxidation indeed occurred within the paddy soil matrix. Over a 48-hour oxidation period, 35.45% of the initial MeHg was consumed in the ≡Fe_3O_4-Fe(II) system and 27.22% in the FeS system (Fig. S11). Furthermore, total Hg (THg) concentrations across all treatments remained stable, fluctuating between 0.21 ± 0.02 mg kg^{-1} and 0.28 ± 0.02 mg kg^{-1} during the incubation (Fig. S10e). Demethylation rate constant (k_d) was further calculated according to Li et al. (2010) and our previous work (Liu et al., 2022; 2023). The maximum k_d in the ≡Fe_3O_4-Fe(II) and FeS systems are 0.022 h^{-1} and 0.192 h^{-1}, respectively (Fig. S11). Notably, the overall k_d values in the FeS system were higher than those in the ≡Fe_3O_4-Fe(II) system, indicating a more rapid demethylation of MeHg driven by FeS oxidation. Moreover, the k_d values during FeS oxidation exhibited an initial sharp decrease before stabilizing (Fig. S11b). In contrast, the k_d values during ≡Fe_3O_4-Fe(II) oxidation displayed an initial increase followed by a subsequent decline (Fig. S11a).

Hydroxyl radical mediated abiotic demethylation of MeHg has been widely documented in aqueous environments with a function of reactive oxygen species (ROS) in photochemical processes (Gårdfeldt et al., 2001; Hammerschmidt et al., 2010; Han et al., 2017; Zhong et al., 2024; Bravo et al., 2026). Model estimation indicated that this MeHg demethylation pathway accounted for 82% of the decline in MeHg concentration in paddy water and 11% of the decline in MeHg concentration in rice around China (Zhong et al., 2024). Recently, Tang et al. (2025) identified a novel, light-independent MeHg demethylation pathway in rice plants mediated by *in vivo* generated singlet oxygen ($^1O_2$). However, the light-impenetrable paddy soil serves as a hotspot for both *in situ* Hg methylation and MeHg uptake by rice roots (Zhao et al., 2016a; Liu et al., 2021). MeHg demethylation in this environment likely exerts substantial control over net MeHg accumulation and subsequent transfer to rice (Meng et al., 2011). Xie et al. (2024) firstly reported a dark abiotic demethylation of MeHg into Hg(0) in Fe(II) minerals containing aqueous conditions. This study demonstrates that abiotic MeHg demethylation mediated by ·OH also occurs in rice paddy soils. In the absence of sunlight, oxidation of Fe(II) in paddy soils under alternated redox drives ·OH generation, which in turn facilitates MeHg demethylation. Similar mechanisms were reported in As(III) oxidation (Meng et al., 2024), Sb (III) oxidation (Tan et al., 2026), Cd mobilization (Zhang et al., 2025), phenol degradation (Chen et al., 2022; Yu et al., 2025), soil organic caron mineralization and CO_2 emission (Liu et al., 2024; Merino et al., 2025) in paddy systems.

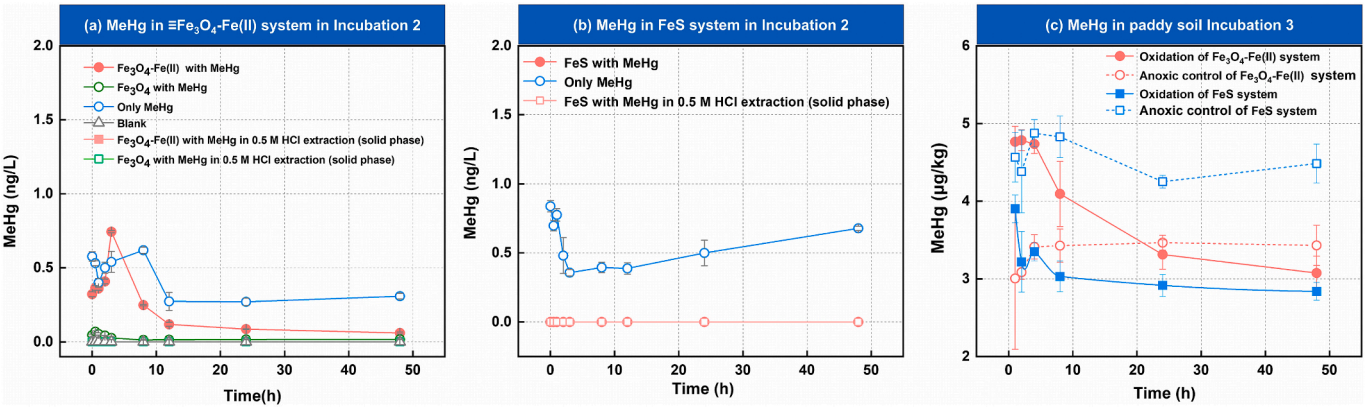


Fig. 4. Concentration of MeHg in both batch incubations (Incubation 2) and paddy soil incubations (Incubation 3). Panels (a) and (b) are MeHg concentrations in Incubation 2 for ≡Fe_3O_4-Fe(II) and FeS systems, respectively. Panel (c) are MeHg concentrations in Incubation 3. Noted that MeHg concentrations in solid phases in batch incubations are reported on a volumetric basis with a fixed extraction agent volume of 20 mL due to the varied mass of solid. The corresponding total Hg concentrations in these incubations are shown in the Fig. S10 in the Supporting Information.

3.4. Divergent mechanisms of MeHg demethylation driven by mineral-specific interfacial electron transfer

The contrasting MeHg demethylation kinetics (Fig. S11) between ≡Fe_3O_4-Fe(II) and FeS imply divergent interfacial electron transfer mechanisms during continuous air exposure. In the ≡Fe_3O_4-Fe(II) system, the generation of ·OH followed a pseudo-first-order kinetic model (R^2 = 0.996, Fig. S5a), indicating that the reaction was predominantly limited by the density of available active Fe(II) sites (Kwan and Voelker, 2003). This single-site controlled oxidation aligned with the observed initial increase and subsequent decline in the demethylation rate constant (k_d) of MeHg (Fig. S11a). During the initial oxidation phase, the reactive surface-adsorbed Fe(II) act as primary electron donor, initiating a successive single-electron transfer cascade that reduces O_2 to ·OH via $O_2^{\cdot-}$ and H_2O_2 intermediates (Meng et al., 2024). However, the Mössbauer spectra demonstrated that the structural Fe(II)(from 15.06 to 13.06%, Table S1) and Fe(III) (from 28.35% to 28.94%, Table S1) within the bulk magnetite core remained shielded. This suggests that almost all

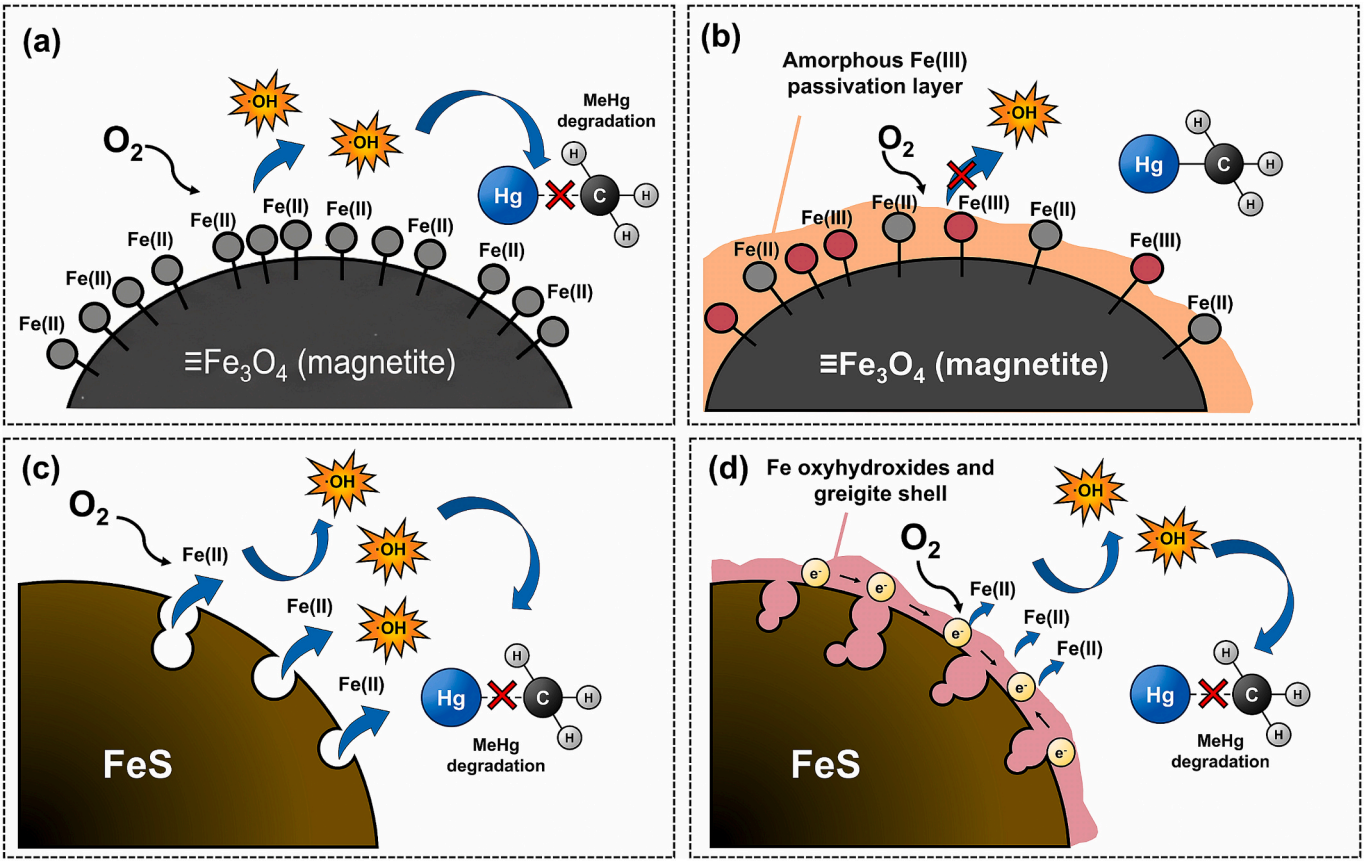


Fig. 5. Illustration for MeHg demethylation driven by mineral-specific interfacial electron transfer. Panels (a) and (b) are early and late periods of ≡Fe_3O_4-Fe(II) oxidation, respectively. Panels (c) and (d) are early and late periods of FeS oxidation, respectively.

electron transfer are derived from the adsorbed species. As the surface Fe (II) was consumed, an amorphous Fe(III) passivation layer formed, hindering further electron transfer. The spectroscopic evidence for this passivation is the decreased relative abundance of surface-adsorbed Fe (II) (from 22.5% to 17.7%, Table S1), alongside potential secondary Fe (III) precipitation as indicated by the QS value (Table S1) (Notini et al., 2023). Consequently, the MeHg demethylation capacity decayed in the later stages of oxidation (Fig. 5a and b).

Conversely, the MeHg demethylation in the FeS system exhibited an initial sharp decrease in k_d before stabilizing (Fig. S11b), which can be explained by the pseudo-second-order kinetic behavior of ·OH generation (R^2 = 0.993, Fig. S5b). Unlike the shielded magnetite core, the structural Fe(II) in FeS actively participates in the oxidation. Initially, the destruction of the FeS lattice released high-spin surface Fe(II), and the synchronous multi-element oxidation of both Fe(II) and sulfide evidenced by S2p XPS (Fig. S4) triggered an initial burst of ·OH (Cheng et al., 2016) and rapid MeHg demethylation. As the initial high-reactivity surface was depleted, the k_d stabilized rather than ceased. This sustained reactivity is likely attributed to an *in situ* generated core–shell structure, such as intermediate Fe_3S_4 ($4FeS + 0.5O_2 + 2H^+ \rightarrow Fe_3S_4 + Fe^{2+} + H_2O$) (Burton et al., 2011; Mu et al., 2025) and Fe oxyhydroxides (Cheng et al., 2020a). Different with oxidized Fe oxides, the newly formed Fe_3S_4 shell possesses electrical conductivity and redox-active sites (Mu et al., 2025). This continuous internal electron pumping regenerates reactive Fe(II) species on the mineral surface, maintaining a steady ·OH production rate (Fig. 5c and d) and explaining the higher overall k_d observed in the FeS system compared to the ≡Fe_3O_4-Fe(II) system (Fig. S11). In addition, demethylation mediated by sulfide may also contribute to the MeHg degradation in the FeS systems (Craig and Bartlett, 1978; Kanzler et al., 2018).

Ultimately, these divergent interfacial electron transfer mechanisms drive a decoupling of reaction kinetics and overall demethylation capacity. While the FeS system generated a larger total pool of ·OH (Fig. 3 and S9) and exhibited a significantly higher initial demethylation rate constant (Fig. S11), it paradoxically achieved a lower net MeHg degradation percentage than the ≡Fe_3O_4-Fe(II) system. This apparent contradiction arises because the rapid ·OH burst in the FeS system suffers from low radical utilization efficiency. In the highly competitive paddy soil matrix, this intense radical flux is likely consumed by non-target scavengers, including the FeS itself (Cheng et al., 2016; 2020b). Conversely, ·OH generation in the ≡Fe_3O_4-Fe(II) system is regulated by a limited density of active surface sites, which dictates a steady and sustained radical release (Kwan and Voelker, 2003). By reducing burst self-quenching and directing steady-state ·OH flux to MeHg over 48-hours, the ≡Fe_3O_4-Fe(II) system ultimately achieves a higher degradation capacity of 35.45% when compared with the FeS system.

3.5. Geochemical implications

Although a specific rice paddy soil was utilized as a model system in this study to demonstrate the dark abiotic demethylation of MeHg, the geochemical implications of these findings extend beyond agricultural settings. Redox-sensitive ferrous iron species are ubiquitous across a wide spectrum of natural redox-dynamic environments, such as tidal marshes, seasonal floodplains, peatlands, and estuarine sediments (Rickard and Luther, 2007; Huang et al., 2021; Dong et al., 2023). In these critical zones, episodic oxygen intrusion, driven by tidal pumping, seasonal hydrological shifts, or macrobenthic bioturbation, can universally trigger the interfacial Fe(II) oxidation and subsequent ·OH burst observed in our experiments. Consequently, this dark abiotic MeHg sink represents a fundamental, yet previously overlooked, component of the global Hg biogeochemical cycle. It operates continuously in light-impenetrable subsurface matrices, suggesting that current Hg transport models, including GTMM model (Smith-Downey et al., 2010) and SiBCASA model (Schaefer et al., 2020) in terrestrial systems, and RPBM model (Kwon et al., 2018) and RMTTM model (Cui et al., 2022) in paddy systems, relying predominantly on photolytic and microbial demethylation likely systematically underestimate the total MeHg attenuation capacity of natural soils and sediments.

Furthermore, the identification of this ·OH-driven abiotic demethylation pathway necessitates a critical re-evaluation of current Hg isotopic fractionation models. Historically, distinct mass-independent fractionation (MIF) signatures, particularly the odd-MIF (Δ^{199}Hg and Δ^{201}Hg), observed in environmental MeHg pools have been attributed to photolytic demethylation driven by light-induced radical pair mechanisms (Bergquist and Blum, 2007; Chandan et al., 2015). Although recent studies have demonstrated that dark abiotic redox reactions (e.g., Hg(II) reduction by Fe(II)) can induce significant kinetic mass-dependent fractionation (MDF) and minor MIF (Schwab et al., 2023), the isotopic consequences of dark, radical-mediated (e.g., ·OH) MeHg cleavage remain unconstrained. By using endmembers of photoreduction, microbial reduction, abiotic dark reduction, our previous work (Zhang et al., 2024) revealed the contributions of each endmember on Hg^0 emission. However, this abiotic dark process was mainly mediated by DOM without the consideration of Fe(II) oxidation mediated ·OH generation. Given the reactive nature of ·OH generated from interfacial Fe(II) oxidation, it is highly plausible that this dark Fenton-like pathway could impart distinct kinetic isotopic signatures.

Finally, the abiotic demethylation of MeHg elucidated here must be contextualized within the broader framework of multi-element biogeochemical coupling. The ·OH generated during interfacial Fe(II) oxidation is reactive and non-selective; similar to its role in MeHg cleavage, this ROS-driven cascade concurrently destabilizes mineral organic matter associations (MOAs) and accelerates the abiotic degradation of protected soil organic carbon (Dong et al., 2023; Liu et al., 2024). This oxidative process also stimulates greenhouse gases emissions, including CH_4 (Liu et al., 2026), N_2O (Yao et al., 2024), and CO_2 (Yao et al., 2025). This presents a profound biogeochemical trade-off: the Fe(II)-mediated Fenton-like reactions that detoxify MeHg may simultaneously accelerate soil organic carbon turnover and compromise carbon sequestration efforts. Collectively, these interconnected pathways underscore an urgent need to investigate the coupled biogeochemical cycling of Hg and key biogenic elements (e.g., C, N, S, and Fe). Addressing these complex interactions is essential to ensure that MeHg mitigation efforts align synergistically with broader climate-smart agricultural goals (Li et al., 2025).

3.6. Limitations

While this study highlights the significant role of Fe(II) oxidation in ·OH production, other pathways likely contribute to dark ·OH generation in natural paddy soils. During redox fluctuations, the oxygenation of reduced soil organic matter (SOM) and other transition metals (e.g., Mn(II)), as well as SOM-metal interactions, can also induce the production of reactive oxygen species (Wang et al., 2022; Yu et al., 2022; Chen et al., 2026; Zeng et al., 2026). Moreover, the oxidation of Fe(II) species at neutral pH can generate non-radical reactive intermediates, such as Fe(IV) (Morin et al., 2023; Bourbon et al., 2024; Yu et al., 2024; Zhang et al., 2026). Although the present study focuses on the role of non-selective ·OH, co-existing Fe(IV), along with other ROS such as superoxide radicals ($O_2^{\cdot-}$), may also contribute to the dynamics of MeHg. More studies in distinguishing and quantifying the relative contributions of ·OH, Fe(IV), and other ROS to the dark abiotic MeHg demethylation are necessary. Furthermore, natural paddy soils contain abundant ·OH scavengers, including dissolved organic matter and ubiquitous inorganic anions (e.g., Cl^- and CO_3^{2-}) (Dong et al., 2023; Zhang et al., 2025). Consequently, the net generation of ·OH and subsequent MeHg demethylation efficiency in natural environments are highly variable and dependent on site-specific soil properties. Therefore, the kinetic parameters (i.e., ·OH yields and MeHg degradation rate constants) derived from our batch incubations cannot be directly extrapolated to field-scale paddy environments. Methodologically, this study utilized controlled Fe

(II) species amendments in pre-autoclaved soils, rather than unamended natural soils, to isolate abiotic degradation pathways. This is attributed to the current methodological challenges, as conventional sterilization techniques inevitably cause premature oxidation of native Fe(II) and thereby hindering the observation of subsequent Fe(II)-driven ·OH generation. Future studies employing non-disruptive microbial inhibition techniques or high-resolution *in situ* stable isotope tracing are required to directly quantify the contribution of native Fe(II) pools to dark abiotic MeHg demethylation in natural field-scale paddy profiles.

4. Conclusions

This study identifies a novel dark abiotic pathway for MeHg demethylation in rice paddy soils, driven by ·OH generated during the oxygenation of reactive ferrous iron species. Air exposure triggered substantial ·OH production from the oxidation of ≡Fe_3O_4-Fe(II)) and FeS, while FeS_2 remained inert under the current experimental conditions. Within 48 h, these Fenton-like reactions facilitated the rapid demethylation of 35.45% and 27.22% of spiked MeHg in autoclaved paddy soil microcosms amended with ≡Fe_3O_4-Fe(II) and FeS, respectively. Mechanistically, reactivity in the ≡Fe_3O_4-Fe(II) system is constrained by available surface sites, whereas FeS undergoes synchronous multi-element oxidation, forming a redox-active core–shell structure that sustains further degradation. These findings fundamentally advance our understanding of Hg biogeochemistry in redox-dynamic environments, demonstrating that models relying solely on microbial and photolytic demethylation likely underestimate the total MeHg sink in light-impenetrable soils.

Data availability

Data are available through the Mendeley Data at https://doi.org/10.17632/tc3yxxpfbf.1 with license CC BY 4.0.

CRediT authorship contribution statement

Jiang Liu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Han Li:** Writing – original draft, Investigation, Data curation. **Chunmei Li:** Writing – original draft, Investigation. **Meng Qi:** Writing – review & editing, Formal analysis. **Sihua Zhu:** Writing – review & editing. **Rong Huang:** Writing – review & editing, Funding acquisition. **Guangdeng Chen:** Writing – review & editing. **Bo Meng:** Writing – review & editing, Conceptualization, Funding acquisition. **Xinbin Feng:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.gca.2026.09.027.

The supplementary material contains one text, one table (Table S1) and 11 figures (Figs. S1-S11). The text describes key parameters of Mössbauer spectroscopy. Table S1 shows Mössbauer spectroscopic measurements on samples at 295 K. Fig. S1 is the XRD and SEM-EDS characterization of the prepared ferrous species. Fig. S2 shows schematic diagram of the experimental design. Fig. S3 shows concentration of solid phases total Fe during ≡Fe_3O_4-Fe(II), FeS, and FeS_2 oxidation in Incubation 1. Fig. S4 is the S2p XPS spectra of FeS before and after oxidation. Fig. S5 is kinetics fitting of ·OH generation. Fig. S6 shows cumulative production of ·OH *v.s.* 0.5 M HCl extractable Fe(II). Fig. S7 is the Fe(II) and Fe(III) dynamics during ≡Fe_3O_4-Fe(II) and FeS oxidation in aqueous and solid phases in Incubation 2. Fig. S8 shows solid phases total Fe during ≡Fe_3O_4-Fe(II) and FeS oxidation in Incubation 2. Fig. S9 shows production of ·OH in Incubation 2. Fig. S10 is the concentration of total Hg. Fig. S11 shows degradation rate constant (k_d) and degradation rate (%) of MeHg.

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